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Shape evolution and capsize dynamics of melting ice

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Ice melting is a complex, dynamic, and multiscale process involving many feedback effects. An essential but poorly understood factor for floating ice is the potential to overturn, as dramatically demonstrated by capsizing icebergs. Here we study the melting-induced reshaping of ice coupled to its free rotation dynamics and find that periods of stable equilibrium are punctuated by spontaneous capsize events. Laboratory experiments on floating ice show that melting produces faceted and angular shapes whose tendency to orient to the waterline is repeatedly upset by rolling over. Similar geometries and motions are reproduced in simulations that drive the shape development via differential melt rate over the surface and which solve for the rotational dynamics subject to shape-dependent forcing from gravity, buoyancy, and hydrodynamics. Melting gradually drives the structure toward postural instability while also priming other orientations that prove stable when reached by abrupt flipping. These findings show how the shape and motions of ice are inextricably linked at fundamental scales, and they reveal mechanisms that may be operative for natural conditions.

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I. INTRODUCTION

The melting of ice and changes in its global levels and distribution are symptoms of our planet's health. Understanding the dynamics of ice is challenging due to the multitude of contributing processes and their interacting effects across space and time. Such complexities arise even at the local scales far below the resolution of weather and climate models. An ice-water boundary recedes in the presence of coevolving flow, temperature, and density fields that are coupled during the phase change, resulting in developments of the interface geometry that in turn feed back on the melting process [1–7]. The scale ranges are vast, with flows of the surrounding water typically being fast and finely structured into thin boundary layers or turbulent plumes while the shape evolves gradually into features and patterns of varying sizes [8–13]. Disentangling such effects at fundamental scales and determining their consequences can inform on the modules that compose global climate models, whose predictive power is limited by subscale uncertainties [14–18].

Idealized settings in laboratory experiments, numerical simulations, and mathematical models complement field studies by providing increased access to the relevant physics and control over the conditions. Such approaches can interrogate single factors in isolation and how multiple effects play off one another in situations that are not naturalistic but nonetheless informative. The canonical context for melting is the so-called Stefan moving-boundary problem that addresses the one-dimensional recession of an interface driven by thermal diffusion in the liquid [19,20]. However, cases of practical interest are intrinsically higher dimensional due to the flows driven by gravitational

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or buoyancy differences. The fluid mechanics of convection is classically studied in systems with fixed and static boundaries, notably the Rayleigh-Bénard configuration of a fluid layer heated from below where instabilities lead to circulation, plumes, and rolls [21,22]. When ice forms one of the solid boundaries, its surface may be patterned by local flows [4,23–26], and additional complexities arise in the double-diffusive convection problem pertaining to melting in saline environments [5,27–29]. Only very recently have the spatial variations in melting and associated changes in shape been investigated for surfaces and bodies of ice under various ambient conditions [3–7,13,30–33]. The geometry feeds back strongly on the flow state and melt rate, and the long-time behavior is therefore not readily forecast from knowledge of the constituent phenomena.

The alarming depletion of the Earth's ice reserves occurs predominantly through melting of floating ice which may move at the free surface, drift, break up, rotate, and overturn [34,35]. Capsizing icebergs are dramatic examples that raise questions about how such motions and the presence of the waterline couple to the many other factors that have been studied for grounded and fully immersed conditions. Previous work on capsizing has emphasized its far-reaching consequences, including changes in global-scale ice dispersal, oceanic flows due to deposition of fresh water, and enormous energy release that may trigger destructive waves and induce collective capsize events [14,15,36–38]. Capsize events are thought to be triggered by changes in shape due to melting and ablation that may undermine postural stability as well as forcing from waves, currents and wind [16,39–41]. Contrarily, it has been argued that melting may have a stabilizing effect [42], and direct confirmation one way or the other from field measurements is missing and perhaps prohibitive. Past studies have sought approximate metrics (e.g., aspect ratio) for gravitational stability by assessing idealized shapes [36,37,41], although other works predict high sensitivity to geometric variations [16,39,40,43]. Such analyses leave unaddressed how an unstable shape might come to be and how capsizing affects the subsequent melting and shape development.

Here we study the dynamics of floating ice as a fully interactive dynamical system involving melting, the consequent shape evolution driven by convective flows, and couplings with the rotational motions via gravitational, buoyancy, and hydrodynamic forcing. We propose and assess an archetypal setting for the capsizing problem in which cylindrical pieces of ice are free to rotate about their symmetry axis while melting in fresh water. Complementary and concurrent work is considering melting and capsizing ice in salt water [33]. Our laboratory experiments allow for characterization of the flows and precise measurements of the shapes and motions through time. A reduced model couples the moving boundary problem for the shape progression to gravitational motion, and numerical simulations reproduce and quantify the experimental observations. These investigations are driven by several goals: to develop and cross-validate methods for capsizing that may be useful for related problems in ice dynamics; to provide experimental data sets needed for appraising such methods; to identify key phenomena whose robustness may be tested in related settings and for different conditions; and to determine the main physical mechanisms and assess if and how they generalize and scale.

II. EXPERIMENTAL METHODS

Here we describe the experimental methods, setups and procedures used in our investigations. The key aspects outlined below include the preparation of cylindrical samples of clear ice, experiments for recording the shape and flip motions during melting in fresh water at room temperature, data extraction and processing methods, and separate experiments for imaging the flows.

A. Sample preparation

The ice used for experiments is grown by a directional freezing method that generates samples free of bubbles and other contaminants [32,44]. Water typically freezes from the outside in, trapping bubbles and other impurities that, if present, can disrupt the melting process, generate flows when released into the bulk fluid, and significantly alter the gravitational stability and floating posture of

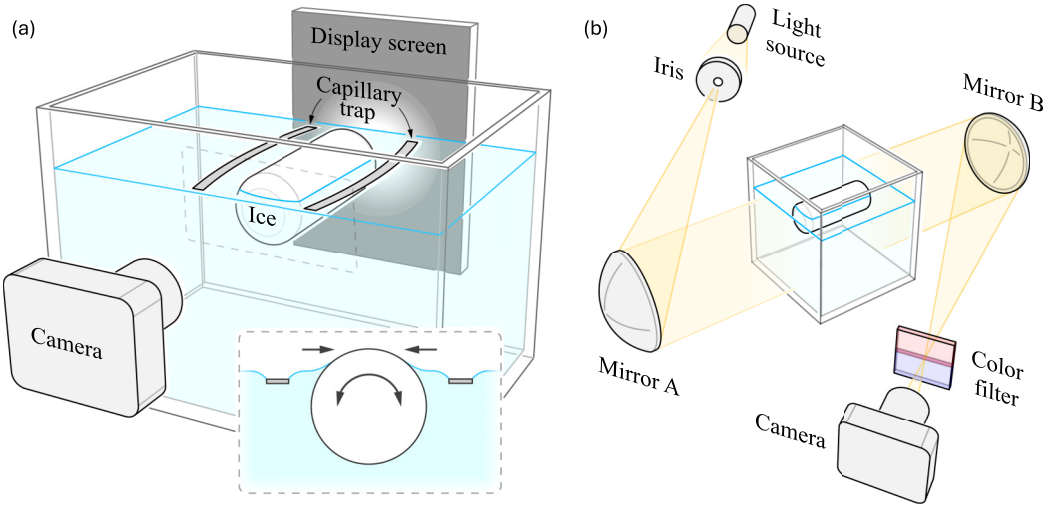


FIG. 1. Experimental setups. The schematics are not to scale, and actual dimensions are provided in the text. (a) System for imaging the shape and motions of ice melting and floating in water. A capillary trap keeps the piece from drifting or rotating out of the view of the camera, and backlighting is provided by a computer-controlled display screen. Inset: Cross-section view showing the operating principle of the trap. The opposite deflections of the menisci on the trap bars and ice lead to stable centering via surface tension interactions. (b) Color schlieren system used for flow visualization. The optical components are arranged in a Z configuration, and differently refracted light transmitted through the sample produce color and intensity variations in the recorded images.

ice. Specialized growth procedures overcome these potential problems by encouraging ice to freeze from the bottom up, thereby allowing bubbles to rise up and away, which is further assisted by continuously stirring the water as it freezes.

Our implementation employs a chest-style freezer at temperature -20°C and in which we include a large bath containing a saturated solution of salt water that remains liquid and which serves as a heat sink. Each sample is grown in a thin-walled, closed-bottom cylindrical mold made of freezer-safe plastic. The mold is filled with filtered and degassed water and then dipped into the freezer and bath, which efficiently conducts heat away and begins bottom-up freezing. The water in the mold is continuously stirred using a gear pump with inlet-outlet tubes inserted through the top, and such agitation serves to wash bubbles and impurities that would otherwise collect at the advancing ice interface. We reduce the flow rate of the pump and lower the mold into the bath in several stages during the growth process. For the molds of diameter 8 cm and height 24 cm used here, it typically takes about 30 h to grow such a sample. These procedures yield clear ice with a small cloudy region at the top where any contaminants are collected and which is cut away after removal from the mold.

B. Apparatus and procedures for measuring shape-capsize dynamics

Our experiments address the melting and capsizing of cylindrical pieces of ice floating in room-temperature fresh water, with the primary goals being to clearly document the cross-section shape and rotational motions with video. The key components of our setup are shown schematically in Fig. 1(a) and involve a large tank of water in which ice is loosely confined within a so-called capillary trap [45] and imaged by a camera with a custom backlighting pattern provided by a display screen [32].

The main considerations that went into the experimental design are maintaining constant far-field temperature of the water, allowing free rotations of the ice while keeping it confined and aligned

in view of a camera, and imaging the interfacial shape of the ice despite it being clear. The tank measures $122 \times 30 \times 40$ cm in length, width and height, which proves to be large enough to keep the far-field temperature at $20 \pm 1^{\circ}$C throughout the 30 min typical run time. The initial shape of the ice is a cylinder measuring 8 cm in diameter and 18 cm in length, and this aspect ratio yields a quasi-two-dimensional system in which flipping occurs only about the axis. Free floating ice would slowly drift and rotate within the plane of the water surface, which we suppress by exploiting surface tension interactions with two bars running along the sides of the ice at several centimeters away. As shown in the inset view of Fig. 1(a), the bars are pressed slightly below the water surface, forming downward-bending menisci that repel the upward menisci on the ice via the Cheerios effect [46]. In this way, the ice tends to stay centered while being free to float and free to roll or capsize along its axis. The bars have a slightly bowed planform shape to restrict drifting forward or backward, and their top surfaces are covered with hydrophobic wax to prevent spillover. Good trapping is maintained as the ice shrinks down by moving the bars incrementally inward over time thanks to an overhead gantry (not shown) with positioning slides.

The ice is imaged along its axis with a digital video camera (Nikon D610) that is precisely fixed at the height of the water surface. An image size of 1280×720 pixels and frame rate of 30 pictures per second is adequate resolution in space and time for the subsequent data analysis. The images are back-lit by a display monitor that illuminates custom-designed intensity patterns generated in MATLAB and which can be dynamically controlled via a computer. We employ a gradient pattern that is brightest in the middle and fades radially outward to black. The difference in the refractive indices of ice and water yields high-contrast boundaries in which the edges of the ice appear black against a lighter background. The position, size, and other parameters of the intensity field are manually adjusted as the ice shrinks to maintain imaging quality.

A typical procedure for an experimental run starts by finely adjusting the water level to be just below the mid-height of the camera so that the two align upon adding the ice. Calibration is done by imaging a ruler. The lighting pattern is displayed and appropriately positioned. The ice is removed from the freezer, allowed to sit for about 15 min to ease its removal from the mold, and then cut to length and its ends polished. The cylinder is then gently lowered within the trap, the bars are cinched together, room lights dimmed, and camera triggered to record. Best results are obtained with two operators, one who monitors and adjusts the backlighting pattern and the other who tunes the trap width.

C. Boundary tracking methods

We extract the cross-sectional shape of the ice from video frames using custom MATLAB scripts. We employ functions such as `imadjust`, `histeq`, and `imsharpen` to preprocess and enhance contrast. To accurately extract the edges of the grayscale image as discrete points, we utilize the Sobel approximation method with a fixed magnitude threshold. Spurious detections associated with the capillary trap bars and waterline are eliminated by gradient-based filtering. We further reduce noise by using `bwareaopen` to remove small clusters. For improved stability and to eliminate overlaid layers, we use `alphaShape` with an optimized `criticalAlpha` parameter. Minor errors are corrected manually using the `ginput` function, which allows for addition and deletion of boundary points. These steps yield a set of 2D coordinates $\{x_i, y_i\}$ that are then interpolated by a Fourier series such that the points are equally spaced in arclength. The number of Fourier modes $N \in \{1, \ldots, 32\}$ is chosen to minimize the mean-squared residual. The scheme leads to some smoothing of sharp features but with negligible effect on the quantities of interest.

D. Apparatus and procedures for flow visualization

The flows are imaged in separate experiments employing the color schlieren apparatus shown in Fig. 1(b), in which slight variations in refractive index due to temperature and density differences are imaged as gradations in color and intensity [47,48]. The principles underlying color schlieren

systems with Z configuration optical assemblies are described in detail elsewhere [47]. Briefly, temperature variations throughout the liquid are associated with nonuniformities in refractive index that differently deflect the transmitted light, and these map to intensity and color variations in the captured images according to the patterns on the filters.

Our system employs a bright lamp (StockerYale Mille Luce M1000) and iris to approximate a point source. A pair of spherical mirrors (Edmund Optics) of diameter 15 cm and focal length 150 cm send collimated light through the sample and condense it through color filters and into the lens of a digital camera (Nikon D610, Sigma 70–300 mm lens). Here the sample consists of an ice cylinder floating in water in a tank measuring 30×30 cm, where this size was chosen to keep as small as possible the interior angles of the Z-shaped path and thereby reduce optical aberrations. We found that color filters of adequate quality can be made by simply drawing stripes on glass slides with standard ink markers, which facilitates the quick testing of different designs. These are placed near the focal point of Mirror B, and their location and orientation greatly affect the imaging quality. The same is true of the camera and other elements in the optical assembly, so meticulous care must be taken in the alignment. A typical experiment is preceded by a lengthy process of fine tuning the optics while imaging ice, for which contaminated and bubbly samples work fine. Once satisfactory, we make use of the previously described methods for manufacturing clear ice and its confinement via capillary trapping, and videos and photographs are recorded.

III. MOVING INTERFACE MODEL

We introduce a reduced order model whose purpose is to account for the coupled shape-capsize dynamical processes and to provide mechanistic insights into the experiments. The essential elements involve a model for the melt rate that may vary in time and with location along the ice surface, and which is coupled to the rigid-body motions of the body under gravitational, buoyancy, and hydrodynamic forces and torques. The experimental conditions motivate the choice of the two-dimensional (2D) setting whose targets for comparison to experiments are the cross-section shape and rotational dynamics. The domain occupied by the ice may have an arbitrary boundary whose evolution is to be solved. The ice floats freely in water and breaches the surface, which remains flat at a fixed height. The ice experiences weight and buoyancy due to gravity, and its translational and rotational motions are resisted by drag forces and torques.

A. Rigid-body dynamics

A schematic of the situation of interest is shown in Fig. 2, which introduces some of the key variables and parameters. Mathematically, the ice domain $\Omega(t)$ lies in the x-z plane with center of mass $(x_{\rm cm}(t), z_{\rm cm}(t))$. The gravitational force $F_g(t)$ and torque $\tau_g(t)$ due to weight and buoyancy can be expressed as integrals over the shape,

$$F_g(t) = g \int_\Omega (\rho(\mathbf{x}) - \rho_{\rm ice})\, dA, \tag{1}$$

$$\tau_g(t) = g \int_\Omega (x - x_{\rm cm})(\rho(\mathbf{x}) - \rho_{\rm ice})\, dA. \tag{2}$$

Here $\rho(\mathbf{x})$ is the density of the background fluid or fluids (here, air and water). The drag force and torque due to motion through the water are assumed to follow established dependencies on size and speed appropriate for moderate to high Reynolds numbers [21,41,49]:

$$F_d(t) = -(C_d \rho_{\rm w} L/\pi) v_{\rm cm} |v_{\rm cm}|, \tag{3}$$

$$\tau_d(t) = -(C_\omega \rho_{\rm w} L^4/\pi) \omega_{\rm cm} |\omega_{\rm cm}|, \tag{4}$$

where C_d and C_ω are drag coefficients, $\rho_{\rm w}$ is the density of water, and $v_{\rm cm}(t)$ and $\omega_{\rm cm}(t)$ are the vertical and angular speeds, respectively. The coefficients $C_d = 10^2$ and $C_\omega = 10^{-2}$ are chosen with

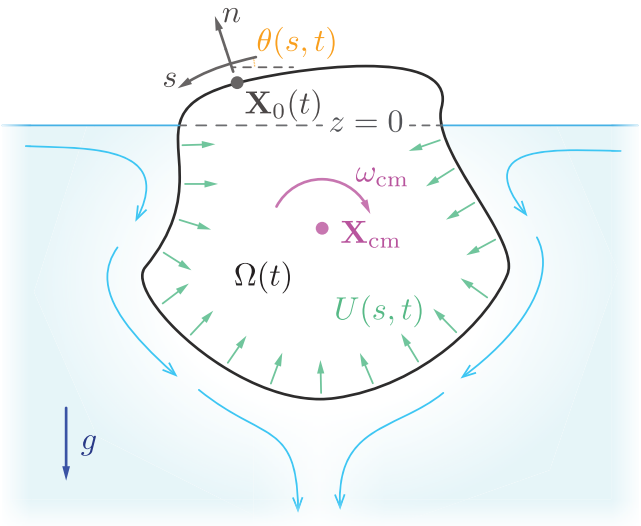


FIG. 2. Schematic of the 2D moving interface model. The ice boundary at time t is represented in terms of its tangent angle $\theta(s,t)$ as a function of arclength location s. An arbitrary reference point $\mathbf{X}_0(t)$, center of mass $\mathbf{X}_{\rm cm}(t)$, and angular velocity $\omega_{\rm cm}(t)$ are useful for describing the dynamics. The melt model assumes a spatially uniform and size dependent rate U for those portions below the waterline ($z=0$), and zero rate above. The body translates and rotates under the action of weight, buoyancy, and hydrodynamic drag.

magnitudes appropriate to circular cylinders at the same size and speed scales [21,22,49,50], and these affect the timescale to equilibrate after a flip. The drag dependence on size is similarly chosen by association with a cylinder of the same perimeter length $L(t)$. The center of mass dynamics then satisfy the Newton-Euler equations

$$M\dot{v}_{\rm cm} = F_g + F_d, \tag{5}$$

$$I\dot{\omega}_{\rm cm} = \tau_g + \tau_d, \tag{6}$$

with mass $M(t) = \int_\Omega \rho_{\rm ice}\, dA$ and moment of inertia $I(t) = \int_\Omega \rho_{\rm ice}[(x-x_{\rm cm})^2 + (z-z_{\rm cm})^2]\, dA$. These equations reflect the quasistatic approximation that, during the rigid-body motions of interest, the boundary recession via melting is much slower so that one can neglect inertial terms associated with $\dot{M}$ and $\dot{I}$. That is, we assume $|\dot{M}v_{\rm cm}| \ll |M\dot{v}_{\rm cm}|$ and $|\dot{I}\omega_{\rm cm}| \ll |I\dot{\omega}_{\rm cm}|$.

B. Model of melt rate

The rigid-body dynamics are coupled to a model of the melt rate $U(s,t)$, which represents the normal velocity of the ice surface at a location specified by the arc length coordinate s. This term drives the boundary motion independent of, and in addition to, the rigid-body motions. The standard law for the recession speed of a melting interface, neglecting temperature variations internal to the ice, is

$$U = \frac{c_p \alpha}{\mathcal{L}} \frac{\partial T}{\partial n} + \varepsilon\kappa, \tag{7}$$

where c_p is the specific heat capacity of liquid water, α is its thermal diffusivity, n is an outward normal coordinate, and $\mathcal{L}$ is the latent heat of fusion [51]. The second term describes enhanced melting at locations of high curvature $\kappa(s,t)$ with the constant coefficient ε dictated by the Gibbs-Thomson effect or other unresolved physics at small scales [30]. The value $\varepsilon = 5\times10^{-4}$ cm^2/s is found to

be a suitable compromise that preserves sharp features of the shape while maintaining numerical stability, and details on the effect of varying this coefficient ε are included in the Appendix.

The general expression given above for the melt rate requires knowledge of the temperature field $T(x, z, t)$ as dictated by the Navier-Stokes and advection-diffusion equations, but can be greatly simplified under the assumptions of stably attached boundary layer flows along a circular boundary of radius R maintained at constant temperature difference relative to the far-field liquid. In such case, the known solution is that the melt rate is uniform over the surface and dependent on the size as $U \propto R^{-1/4}$, where the constant prefactor depends on the temperature difference and fluid material parameters [22]. Our model assumes this solution for the wetted portions of the ice boundary below the waterline, where the effective radius $R = L/2\pi$ relates to the perimeter $L(t)$. The melt rate for those portions exposed to air is taken to be zero. The abrupt change in rate at the waterline presents numerical instabilities that are mitigated by smoothing over a length scale set by the capillary length $\lambda = 0.27$ cm for an air-water interface. Mathematically, $U = \beta H(z/\lambda) + \varepsilon\kappa$, where $H(z) = [1 - \tanh(2z)]/2$ is the regularized step function. Using the values for water at $T = 20^{\circ}$C, we find $\beta = \beta_0(L/L_0)^{-1/4}$ with $\beta_0 = 2.9 \times 10^{-3}$ cm/s.

C. θ-L formulation of interface dynamics

Let $\mathbf{X}(s, t)$ be a parametrization of the boundary with arclength s. Each point on the boundary has total velocity due to melting, translation, and rotation,

$$\frac{\partial \mathbf{X}}{\partial t} = U\hat{\mathbf{n}} + V\hat{\mathbf{s}} + \dot{\mathbf{X}}_{\mathrm{cm}} + \omega_{\mathrm{cm}}(\mathbf{X} - \mathbf{X}_{\mathrm{cm}})^{\perp}, \tag{8}$$

where $\hat{\mathbf{n}}(s, t)$ is the normal vector to the curve, $\hat{\mathbf{s}}(s, t)$ is the tangent vector, and $\mathbf{x}^{\perp} = (-y, x)$. The normal speed U defines the local melt rate, and the tangential speed V can be chosen arbitrarily as it does not affect the boundary shape. A reference point $\mathbf{X}_0(t) = \mathbf{X}(0, t)$ serves to track the laboratory-frame motion, and its velocity satisfies $\dot{\mathbf{X}}_0 = U(0, t)\hat{\mathbf{n}}(0, t) + \dot{\mathbf{X}}_{\mathrm{cm}} + \omega_{\mathrm{cm}}(\mathbf{X}_0 - \mathbf{X}_{\mathrm{cm}})^{\perp}$. We evolve the boundary via the so-called θ-L method, which has been shown to admit stable and accurate discretizations in related settings [30,52,53]. The ice boundary is represented in terms of its tangent angle $\theta(s, t)$ and total arclength $L(t)$ so that $\hat{\mathbf{s}} = (\cos\theta, \sin\theta)$, $\hat{\mathbf{n}} = (-\sin\theta, \cos\theta)$, and $\mathbf{X}_s = L\hat{\mathbf{s}}$. Differentiating the representation for $\mathbf{X}_t$ with respect to s and $\mathbf{X}_s$ with respect to t then gives

$$\frac{\partial \theta}{\partial t} = \omega_{\mathrm{cm}} + \frac{1}{L}\left(\frac{\partial U}{\partial s} + V\frac{\partial \theta}{\partial s}\right), \tag{9}$$

$$\frac{dL}{dt} = -\int_0^1 \frac{\partial \theta}{\partial s} U \, ds, \tag{10}$$

$$V = \int_0^s \frac{\partial \theta}{\partial s} U \, ds' - s\int_0^1 \frac{\partial \theta}{\partial s} U ds'. \tag{11}$$

The expression for the tangential velocity V arises from the constraint $|\mathbf{X}_s| = L$ and ensures s remains an arclength parameter [54].

D. Numerical implementation

We solve these equations using a custom MATLAB implementation. The numerical method employs a pseudospectral discretization in $s \in [0, 1)$ of the tangent angle θ and a fourth-order, implicit-explicit multistep method in time. The angle diffusion term $\varepsilon\kappa_s = \varepsilon\theta_{ss}/L$ is treated implicitly while all other terms are explicit. In addition to the tangent angle, the variables evolved are the total arclength $L(t)$, the translational and angular velocities $v_{\mathrm{cm}}(t)$ and $\omega_{\mathrm{cm}}(t)$, and the reference point $\mathbf{X}_0(t)$. The force, torque, mass, center of mass, and moment of inertia are computed from the parametrization $\mathbf{X}(s, t)$ using Green's theorem along with the spectrally accurate trapezoidal rule.

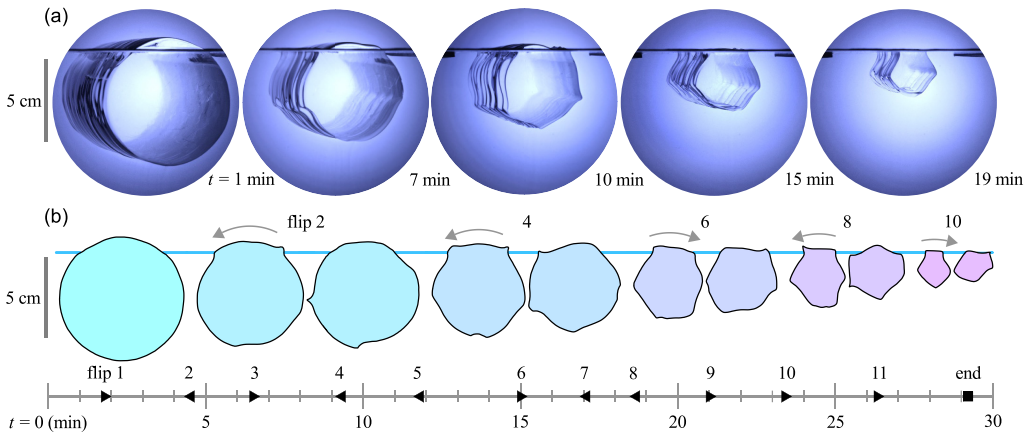


FIG. 3. Shape and capsize dynamics of floating ice cylinders melting in room temperature water. (a) Photographs at indicated times show the development of corners and facets in the cross-sectional shape and quasiperiodic ripples along the length. The ice is static in each snapshot and flips one or more times between the photos. (b) Extracted cross-sectional shapes immediately before and after selected flips. Each flip is marked on the timeline, showing time of occurrence and sense of rotation.

The tangent angle θ will exhibit a jump discontinuity since $\mathbf{X}$ is a closed curve. This can be avoided by writing

$$\theta(s,t) = 2\pi s + \sum_{n=0}^{N-1} \hat{\theta}_n(t) e^{2\pi i n s}, \tag{12}$$

and then evolving the Fourier coefficients $\hat{\theta}_n(t)$, $n = 0, \ldots, N-1$. In all simulations we take $N = 2048$, using the 3/2 anti-aliasing rule to treat nonlinear terms pseudospectrally. To be explicit, writing the set of equations generically as $\dot{\mathbf{u}}(t) = \mathbf{L}\mathbf{u} + \mathbf{f}(\mathbf{u})$, where $\mathbf{u} = (\hat{\theta}_0, \ldots, \hat{\theta}_{N-1}, L, v_{\rm cm}, \omega_{\rm cm}, \mathbf{X}_0)^T$ is the set of all variables, $\mathbf{L}$ is a matrix, and $\mathbf{f}(\mathbf{u})$ is a nonlinear function, each time step takes the form

$$\left(\tfrac{25}{12}\mathbf{I} - \Delta t \mathbf{L}^{n+1}\right)\mathbf{u}^{n+1} = 4\mathbf{u}^n - 3\mathbf{u}^{n-1} + \tfrac{4}{3}\mathbf{u}^{n-2} - \tfrac{1}{4}\mathbf{u}^{n-3} + \Delta t(4\mathbf{f}^n - 6\mathbf{f}^{n-1} + 4\mathbf{f}^{n-2} - \mathbf{f}^{n-3}), \tag{13}$$

where the superscript denotes the value at time $t^n = n\Delta t$. The values at steps $n = 0, 1, 2$ are initialized with a fourth-order Runge-Kutta method. In all cases we take $\Delta t = 0.005$.

IV. RESULTS

Here we present measurements from 10 experimental trials and the corresponding outputs of simulations. This wealth of data provides ample opportunities to cross-validate the methods, compare and contrast results, and qualitatively and quantitatively characterize the dynamical interactions between melting and capsizing. We focus on salient aspects of the problem, including surveys of the shape-rotation motions; experimental visualizations of the flows; assessments of the melt rate along the surface and through time; quantitative characterizations of the shape development; and analysis of the angles of rotation during capsize events as well as their number and distribution over time.

A. Overview of shape-rotation dynamics in experiments

The photographs of Fig. 3(a) show snapshots at the indicated times from a representative experiment, and here a slightly angled view gives a sense of the 3D shape and its development. The images are taken at moments when the ice is static, and it flips one or more times between the

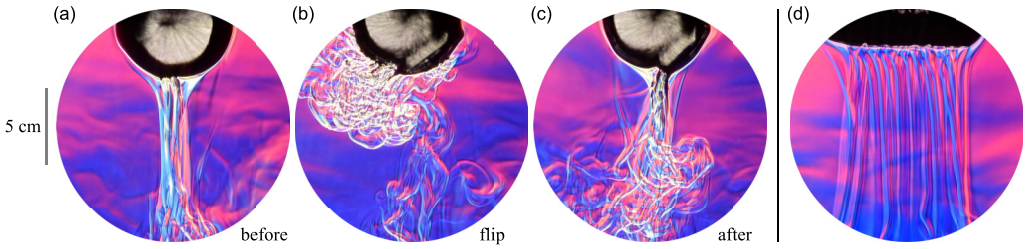


FIG. 4. Flow imaging experiments via the color schlieren method. (a)–(c) Axial view photos just before, during, and after a flip. The boundary layer flows adhere to the lower surface of the ice and detach at the bottom. This state is disrupted during a flip and then reestablished within about 30 s. (d) A side view shows that the detaching flows have periodic variations spaced by about 1 cm along the length of the ice.

images. Corresponding videos provided as Supplemental Material [55] show that the ice undergoes a series of repeated events in which it locks in place for a few minutes while gradually melting and then abruptly rolls over and assumes another static orientation. This process yields 10 to 15 flips during the roughly 30 min it takes to melt away. During each locked orientation, the cross-section develops sharp cornerlike features near the air-water interface. In the 3D view of the images, these are seen as ridges that run the length of the cylinder. The flips bring new portions of the ice surface to the waterline, where corners again develop. These dynamics produce cross-sectional forms with intricate features that at later times yield a polygonlike shape with flat facets separated by sharp corners. The shape variations along the length are comparatively simple, taking the form of centimeter-scale undulations that develop early on and persist throughout the run. We therefore focus on the 2D cross-section shape in most of the following characterizations.

Aligning the camera with the axis of the cylinder yields videos whose images are analyzed to extract the ice boundary shape over time by the methods described in Sec. II. Sample data from a representative trial is shown in Fig. 3(b), where a quasicircular shape at an early time is followed by pairs of shapes just before and after selected flip events. The timeline below charts out the flip events with triangular markers that indicate when they occur and the sense of rotation. The corners that develop at the waterline are clearly seen in the first member of each pair. Rolling over may fully submerge both just-formed corners, e.g., flip number 2, but more commonly one of the two is returned to the waterline on the other side, e.g., flips 4, 6, 8, and 10. These dynamics lead to patterning of corners all around the shape, giving a faceted quasipolygonal geometry at later times. The timeline shows that flips occur every few minutes with some variability, and there seems to be no clear pattern to the direction of rotation. These observations are consistent across 10 trials, three of which are visually surveyed in the Appendix. All data sets are provided in the repository specified there.

B. Experimental flow visualizations

Preliminary insights into the mechanisms come from visualizing the flows of water near the ice. The photographs of Fig. 4 are obtained in separate experiments that employ the color schlieren method, as described in Sec. II. Figures 4(a)–4(c) provide axial views just before, during, and soon after a flip, and Fig. 4(d) shows a side view when the ice is locked. Corresponding videos are provided as Supplemental Material [55]. The locked state visualized in Fig. 4(a) is characterized by flows of the cold, dense water near the ice that stream downward and conform to the surface until separating near the lowest point. These observations are consistent with attached flows in boundary layers driven by density differences in the liquid. This condition is momentarily disrupted in Fig. 4(b) during a flip as the flow detaches and is highly unsteady. Similarly complex flows are

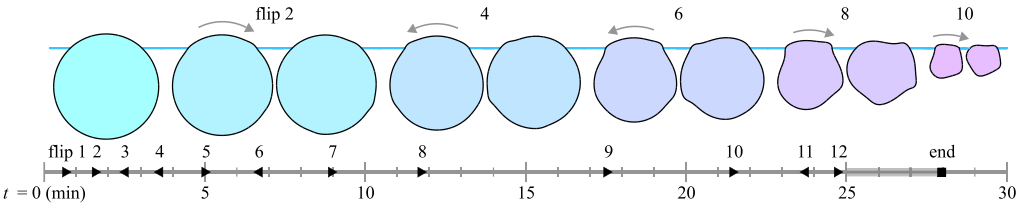


FIG. 5. Boundary shapes and flip dynamics computed in numerical simulations. The initial shape is a circle, and subsequent shapes are shown before and after selected flips. Flip occurrences and sense of rotation are marked on the timeline. The last several minutes are marked with a gray band and involve a rounded shape that rolls continuously.

observed during the capsizing of nonmelting objects [37]. The boundary layer flow with detachment point below is reestablished in Fig. 4(c), which is about half a minute after the flip.

In the locked state, the detaching flows are structured into an array of plumes running the length of the ice, as shown in Fig. 4(d). This flow structure is consistent with Rayleigh-Bénard convection [21], for which detachment via regularly spaced plumes reflects the periodic route by which the flow goes unstable. The periodicity of the flow is also imprinted on the ice as the ripples along its length seen in the photos of Fig. 3(a) and which manifest in related settings involving gravitationally unstable flows induced by melting or dissolving surfaces [4,24,26,56].

C. Shape-rotation dynamics in simulations

We next survey the numerical results in simulations that start with a circle of diameter 8 cm and then evolve the shape and rotation according to the methods of Sec. III. Figure 5 charts out the computed ice boundary shapes before and after selected flips, all of which are charted out on the timeline below. A corresponding animation is provided as a Supplemental Video [55]. Similar to the experiments, the ice undergoes repeated lock-and-flip motions through which it develops a faceted geometry. Corners are formed at the waterline during locked states, and repeated flips pattern these features around the perimeter. In comparison to the experiments, the corners of the simulated boundaries are generally smoother, which can be attributed to the regularizing effects in the melt law Eq. (7) associated with the transition at the waterline and the curvature term. These effects are also likely responsible for the rounded shapes seen at very late times in the Supplemental Video [55] and which lead to continuous rolling during the stage marked with a gray band in the timeline of Fig. 5. The effects of the curvature term are explored further in the Appendix. Despite the differences between experiments and simulations, the general aspects and trends seem to be well reproduced at a qualitative level. So too are many quantitative aspects such as the flip count of 12 and total time to melt of 28 min, suggesting that the model contains much of the essential physics.

D. Melt rate

Inspection of the experimentally measured boundaries during locked states indicates that the melt rate is nonuniform along the ice surface. A representative trial is shown in Fig. 6, where six locked states are assessed by overlaying their beginning (lighter shade) and ending (darker) shapes. The selected states and their duration are marked on the timeline below. Close examination shows that inferring the local melt rate is complicated by the fact that the ice is not truly static during a locked state but rather undergoes small translations and rotations. For cases in which these motions are especially small, three trends are apparent. First, the melting rate for the portion of the surface exposed to air can be seen to be significantly lower than that of the wetted surface. Second, the melt rates tend be higher when the ice is smaller at later times. Third, the melting rate on the wetted portion varies with location and specifically seems to be somewhat higher for those locations nearer

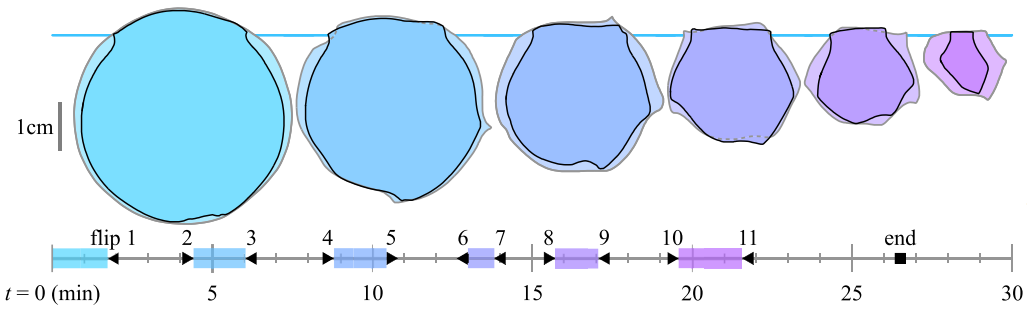


FIG. 6. Ice boundary motion during locked states in a representative experiment. Every other locked state is documented as indicated on the timeline, and each is represented by overlaying the shapes from the beginning (gray boundary and lighter fill color) and end (black boundary and darker color) of the state.

the waterline than those further down. Our melting model incorporates the first two effects but neglects the third by assuming uniform rate along the wetted surface.

The overall rate of melting can be assessed quantitatively by examining the cross-sectional area $A(t)$ of the ice, which is seen in experiments and simulations to decrease continuously from its initial value A_0 and eventually vanish at a final time t_f. This motivates the dimensionless characterization of A/A_0 versus t/t_f, which is plotted in Fig. 7(a) for 10 experimental trials (filled cyan and magenta markers) and the simulation (open blue markers). Data points for experimental trials are taken immediately following each capsize event, and data points for the simulation are displayed once per minute. Plotting the data in this nondimensional way uses the measured A_0 at an early time and an inferred value of t_f determined by fitting a curve to $A(t)$ and finding its zero crossing. The results show strong agreement between experiments and simulation, all of which follow the same nonlinear trajectory. Moreover, the shrinking rate proves to be consistent with a power law decay of the form $A/A_0 = (1 - t/t_f)^{8/5}$, which is plotted as the black dashed curve. These results are verified on the log-log plot of the inset, where the data all fall along a line of slope $8/5$. The origin of this particular power will be discussed in Sec. V.

E. Modal analysis of the shape development

We now take up the more nuanced issue of the shape progression, which shows a tendency in experiments and simulations toward a shape with distinct facets and corners. This motivates a metric that seeks to assign a number of edges that is most consistent with a given shape. Our method considers the curvature $\kappa(s)$ as a function of the dimensionless arclength $s \in [0, 1]$ around the shape, and the example plot from experiments in the inset of Fig. 7(b) shows how cornerlike features of the shape correspond to peaks in $\kappa(s)$. The spacing between such peaks suggests a dominant spatial frequency, which is confirmed by the identification of the wavenumber k^* that maximizes the power spectrum of the Fourier transform of $\kappa(s)$, as described in more detail below. The so-called shape mode k^* determined in this way is an integer that can be interpreted as the number of corners that can fit around the shape, implying an association with a polygon of k^* sides. For example, $k^* = 5$ indicates that sharp features are spaced by about 1/5 of the perimeter length, as for a pentagon.

We assess the shape mode across time for all 10 experimental trials and the simulation, and the results are summarized in the main plot of Fig. 7(b). The experimental markers represent the shapes just after each flip, and simulation points are shown every minute. These data sets show general agreement and most strikingly reveal the convergence to a plateau at the dominant mode value of $k^* = 5$. The mode numbers tend to be larger and variable at early times, which reflects rough shapes without consistent structure across trials. By 5 to 10 min and after the first few flips, the $k^* = 5$ mode is selected and thereafter persists until the very latest times just before vanishing.

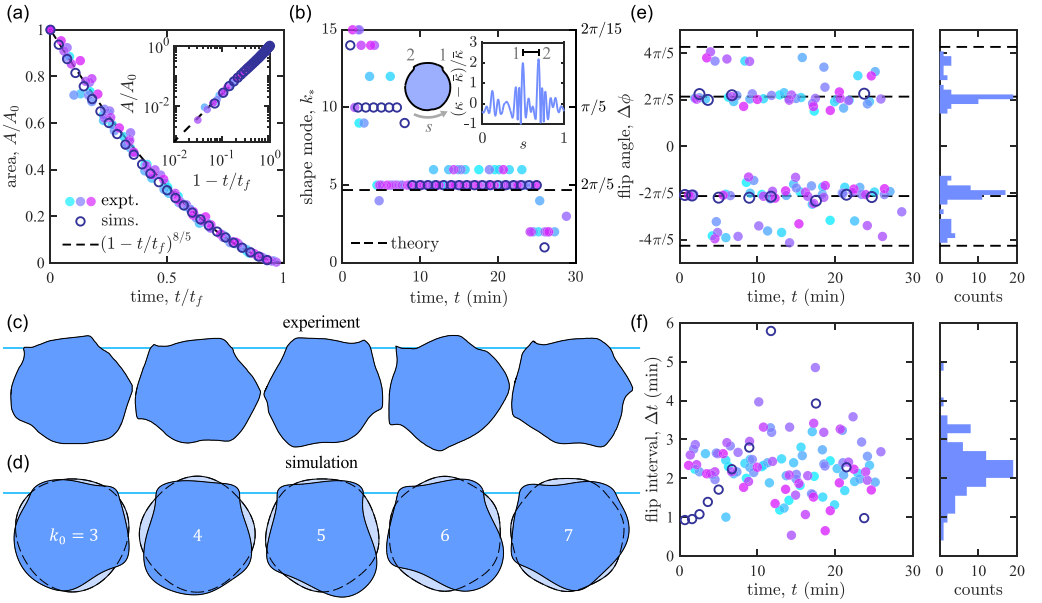


FIG. 7. Shape dynamics over 10 experiments (solid markers) and a simulation (open markers) with initial shape of a circle. (a) Cross-sectional area of the ice shrinks according to an 8/5-power law consistent with boundary layer theory scaling. (b) Evolution of the shape mode shows convergence toward a quasipentagonal geometry with $k^* = 5$. The dashed black line marks the value 4.7 predicted by a geometric model. Inset: an example shape with corners and corresponding peaks in its curvature versus arclength function from which the mode is extracted. (c) Ice boundary shapes taken at the midpoints of 5 experimental trials show a common quasipentagonal form. (d) Computed boundaries for initial shapes (pale blue) perturbed with the indicated mode k_0. The later shapes (dark blue) at the midpoint of each run are overlaid. (e) Rotation angles of capsize events tend to cluster near nonzero integer multiples of the angle $1.34 = 77^\circ$ (dashed lines) predicted by a geometric model. (f) The duration of locked states between flips is typically 2 to 3 min with considerable variability.

These results show that, despite variations between trials and differences between experiment and simulation, there exists a strong statistical consistency in the shape dynamics. The observed shape mode selection is a primary target for the explanatory mechanisms presented in Sec. V.

To more fully explain the data processing steps behind the shape mode determination, we present the analysis of a representative experiment in Fig. 8. The plots of Figs. 8(a) and 8(b) address three chosen times t whose values are reported in Fig. 8(a) and for which the shapes are given as insets in Fig. 8(b). In Fig. 8(a), we plot the curvature κ versus arclength s, both quantities normalized to facilitate comparison across different times and sizes. The quantity $(\kappa - \bar{\kappa})/\bar{\kappa}$ uses the mean curvature $\bar{\kappa}$ to remove the $k = 0$ mode and ensure a consistent relative scale over time, and arclength s is scaled on the perimeter L. The plots show that sharp corners show up as peaks that are marked with numbers and which can be matched to the shapes in Fig. 8(b). We then apply the discrete Fourier transform to arrive at the amplitude a_k for each wave number k: $a_k = \sum_{n=0}^{N-1} \kappa(n\Delta s) e^{-i2\pi kn/N}$, where κ and s are their normalized versions and s is discretized into N segments of size Δs. The corresponding power spectra are plotted in Fig. 8(b), and the dominant mode k^* in each case is determined by the location of the maximum (circled). The early time in the top panel has weak amplitudes across all k, implying a shape without clear mode selection. The middle panel indicates that somewhat later the mode $k^* = 5$ stands out, and the corners are more pronounced. The lower panel shows that $k^* = 5$ is further reinforced later on, and the power is low elsewhere except at the harmonics. Related, the curvature shows five dominant peaks, and the

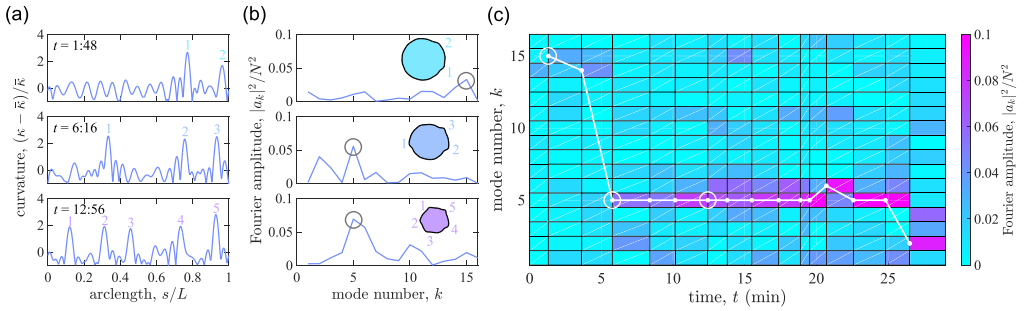


FIG. 8. Shape mode analysis of a representative experiment. (a) Normalized curvature $(\kappa - \bar{\kappa})/\bar{\kappa}$ plotted versus normalized arclength s/L around the boundary at three time points. (b) Corresponding power spectra $|a_k|^2/N^2$. Inset: Boundary shapes at the selected times. (c) Map of power across time and mode number with the dominant mode tracked by the white dots connected by lines. The three selected times from panels (a) and (b) are circled.

shape appears quasipentagonal. A complete analysis for all times is shown in the map of Fig. 8(c), where color indicates the power across time t and wave number k. The example time points from Figs. 8(a) and 8(b) are circled. The values of the dominant mode k^* are shown as white dots joined by lines, and these are the values that go into the plot of Fig. 7(b) for this particular trial. The power is uniformly weak at early times, as expected of featureless shapes. After about 5 min, modes at or near $k = 5$ emerge as dominant and thereafter intensify as the corners are reinforced.

F. Quasipentagonal terminal shape

The selection of the shape mode $k^* = 5$ across all experimental trials and in simulations suggests that a quasipentagonal form can be considered the terminal shape or attractor of the shape dynamics. The robust tendency toward a five-sided shape is corroborated by the sample boundaries displayed in Fig. 7(c), which are selected from near the midpoints of five experimental trials. Such times can be considered indicative of the terminal condition in the sense that the ice has capsized 5 to 10 times, and the corners formed at the waterline therefore appear all around the perimeter. Here we scale the contours to an equivalent area to facilitate comparison of their shape. While varying in their geometry and deviating substantially from regular polygons in all cases, the preference toward 5 sides is readily appreciated. It is also seen that the side length seems to match the intersection with the waterline, which inspires the geometric model given in Sec. V and which produces the prediction shown as the dashed line in Fig. 7(b).

To further test for the existence of a terminal shape and determine its form, we carry out additional simulation trials for initial geometries that are perturbed away from a circle. We consider a starting shape modulated with low-amplitude radial undulations that have mode number k_0 and which therefore impart as many lobes. In Fig. 7(d) we assess $k_0 = 3$ to 7, where the starting shapes are shown in light blue with dashed outlines and the resulting shapes at the midpoint are shown in darker blue with solid outlines. All the shapes are again rescaled to a common area, and select surveys of the time progressions are provided in the Appendix. Compellingly, these trials show that, even when another mode is the only one present at the outset, it is eventually replaced with $k^* = 5$. We therefore view a quasipentagonal form as a stable attractor of the shape dynamics. Shape selection is a feature shared with related problems involving erosion, dissolution, and melting in flow [32,52,57–62].

G. Rotation angle and frequency of capsize events

Finally, we assess the capsize events themselves, starting with the angle through which the ice rotates. In Fig. 7(e) we chart out the flip angle $\Delta\phi$ across all 10 experiments and the simulation.

These measurements involve identifying the time at which a flip occurs based the high angular velocity and then subtracting the orientation angles before and after. Remarkably, the data across experiments and simulations display a strongly preferred angle that is most clearly seen in the histogram of experimental measurements shown to the right. Two narrow and isolated peaks are seen to arise symmetrically about the origin, and weaker and more diffuse clusters are seen at larger angles. Pooling the data in the primary peaks and computing the mean and standard deviation, we identify a preferred angle $1.3 \pm 0.2 = 71^\circ \pm 8^\circ$. This value can be compared with $2\pi/5 = 72^\circ$ as the corner-to-corner interior angle for a pentagon or generally shapes of dominant mode $k^* = 5$. Further, the secondary clusters seem to be centered near twice this angle. These data therefore indicate that the flip angles have a quantized structure in that they tend to come in integer multiples of a fundamental unit.

These results, and the clear connection with the shape mode selection, are targets for a quantitative explanation presented in Sec. V. Qualitatively, the picture is that the ice tends to rotate such that one of the two corners carved at the waterline during the locked state is again returned to the waterline after capsizing. Considering such events repeated and flips of either sense, one can understand how existing corners serve as templates for additional corners at regular spacing and hence how the quasipolygonal form comes to be.

The frequency of capsize events can be quantified by the time interval between flips, i.e., the duration of the locked states. The measured values of the interflip intervals Δt are plotted versus time in Fig. 7(f) for the experiments and simulation. These data are compiled by identifying when each flip occurs based on the high angular velocity. The experimental values show no significant structure in time, and hence we pool all the measurements into the histogram to the right. The mean and standard deviation are 2.3 ± 0.7 min, which corresponds to the observed 10 to 15 flips in the run time of about 30 min. The simulations with a circular initial shape show similar timescales, with short interflip intervals seen both early and late and longer durations at intermediate times. However, this trend is not preserved in additional runs involving modal perturbations to the initial shape or alterations to the strength of the curvature term in the melting law, as detailed in the Appendix. Based on these findings, we conclude that the characteristic timescale between flips is fairly robust but other aspects of their temporal distribution may be sensitive to geometric details.

V. MECHANISMS AND INTERPRETATIONS

The tractability of the model and quantitative validation with experiments suggest that it can be used to investigate in detail the mechanisms underlying the observations and phenomena. In what follows, we provide additional interpretations and analyses of the model outputs that help to disentangle and relate the coupled shape-rotation dynamics. These inform on the melting rate variations, the shape development into a terminal form, the tendency to flip through a characteristic angle, and dynamical instabilities that can be assessed through the gravitational potential and metacenter.

A. Melting rate variations

The melting rate is observed to vary over time and also along the ice surface at a given time. The former effect is quantified by the power law $A(t)/A_0 = (1 - t/t_f)^{8/5}$ describing the temporal decrease in cross-section area and which is shown in Fig. 7(a) to account for the experimental measurements and simulation outputs. The boundary layer theory scaling underlying the model readily explains this result. Namely, the model assumes a melt rate $U \sim L^{-1/4}$ that varies with the length scale L of the ice. The area shrinks at a rate dictated by U along the perimeter length, which together with the geometric relation $A \sim L^2$ yields $\dot{A} \sim UL \sim A^{3/8}$. This separable differential equation is readily solved to yield the the observed 8/5 power law in time. The nontrivial power and its corroboration in simulations and experiments can be taken as quantitative evidence that melting occurs predominantly within boundary layer flows.

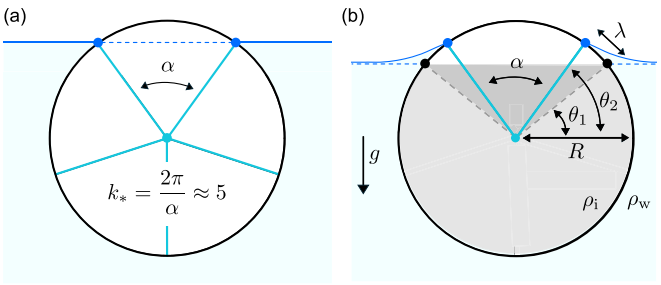


FIG. 9. Geometric model of the faceted shape and related flip angle. (a) The angle α is formed by the intersections of the ice with the waterline and divides the circle into k^* sectors. (b) Quantities that determine α include the gravitational constant g, the ice size R, the ice and water densities ρ_i and ρ_w, and the capillary length λ. The angles θ_1 and θ_2 are useful in calculations.

The dominant spatial variation is associated with the significantly slower melt rate for the upper portion of the ice out of the water as compared to the wetted portion below. Our model assumes no melting of the iceberg tip, which can be justified by the low thermal conductivity of air, the consequently thick thermal boundary layers, and low melt rate via Eq. (7). The experimentally observed melt rate also varies along the wetted surface, with a tendency toward higher rates near the air-water interface (Fig. 6). In contrast, the model assumes a spatially uniform rate as predicted by the boundary layer theory solution for a circular boundary fully immersed in a fluid and with a fixed temperature difference relative to the far field. The departure of the model from experiments can therefore be attributed to the air-water interface, whose effect on the flows is proposed in Fig. 2. In this view, the higher than expected melt rate just below the waterline is due to warm fluid being drawn in along the interface where it feeds into the boundary layer flow. Nevertheless, this factor does not seem essential insofar as the uniform-rate model captures much of the dynamical phenomena and trends seen in experiments.

B. Pentagonal terminal shape and characteristic flip angle

The experimental images and model results motivate a simple geometric explanation for the five-sided terminal shape and the related flip angle through approximately one-fifth of a revolution. We propose that both arise from the ice breaching the air-water surface and the consequent effect on melting. As illustrated in Fig. 9(a), the intersection points of the ice with the waterline determine the spanning angle α and the dominant shape mode $k^* = 2\pi/\alpha$. The physical picture is that the intersections are sites of discontinuity in the melt rate that yield corners during the locked state, and each flip through an angle α returns one of the two just-formed corners to the waterline. Repeated lock-and-flip events then yield a faceted shape with k^* corners.

The angle α is therefore determined by the equilibrium height of the ice in the water, which involves weight and buoyancy but also surface tension at the centimeter scales of our experiments. The latter affects the vertical force balance as well as the meniscus, which rises up along the ice surface a distance comparable to the capillary length $\lambda = \sqrt{\gamma/(\rho_w - \rho_a)g} \approx \sqrt{\gamma/\rho_w g} = 0.27$ cm as determined by the surface tension coefficient $\gamma = 72$ dyn/cm of the air-water interface and their densities [63]. The schematic of Fig. 9(b) defines some useful quantities. Assuming the water to wet the ice with zero contact angle, vertical force balance can be expressed in terms of the angle θ_1:

$$\rho_i \pi R^2 g + 2\gamma \sin\left(\frac{\pi}{2} - \theta_1\right) = \rho_w R^2 \left(\frac{\pi}{2} + \theta_1 + \sin\theta_1 \cos\theta_1\right) g.$$

The three terms represent per-unit-length forces for the weight of the ice, the downward component of the surface tensions from the two contact points, and buoyancy. The latter is associated with the region of the ice below the far-field waterline, whose area is calculated as the sum of a sector of a

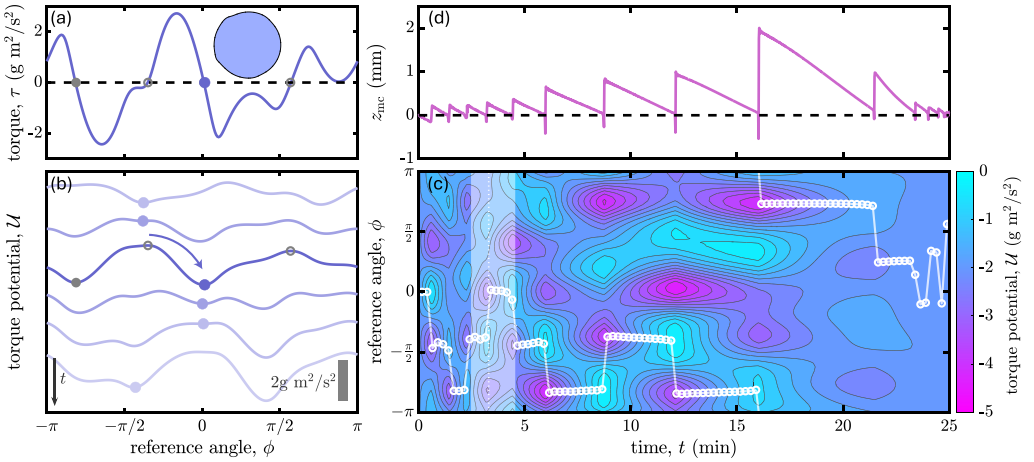


FIG. 10. Transition to instability in simulation. (a) Torque on a representative geometry upon rotating through a reference angle ϕ. Two stable (solid) and unstable (open) equilibria are identified, with the current configuration (purple) being stable. (b) The torque potential $\mathcal{U}(\phi)$ associated with (a) (dark purple curve), including snapshots both forward and backward in time (light purple curves). The ice loses stability through a subcritical pitchfork bifurcation and capsizes. (c) Temporal evolution of the torque potential $\mathcal{U}(\phi, t)$. The dotted line corresponds to the moment shown in panel (a) and the shaded strip indicates the period from which snapshots in panel (b) are selected. (d) The metacentric height z_{mc} decreases in time while in a stably locked state and then abruptly jumps once $z_{mc} = 0$ as the ice capsizes.

circle (lighter gray) and a triangle (darker gray). Given the constants $g = 980$ cm/s^2, $\rho_w = 1$ g/cm^3, and $\rho_i = 0.92$ g/cm^3, and taking the size to be the initial radius $R = R_0 = 3.8$ cm, the above nonlinear algebraic equation is solved numerically to yield $\theta_1 = 0.83 = 48^\circ$. Approximating the wetting arc length of the meniscus to be λ, then $\theta_2 = \theta_1 + \lambda/R$ and $\alpha = \pi - 2\theta_2 = \pi - 2(\theta_1 + \lambda/R) = 1.34 = 77^\circ$. This is close to the value 72° as the corner-to-corner interior angle for a pentagon, and the corresponding shape mode $k^* = 2\pi/\alpha = 4.7$ is close to the number of corners of a pentagon. These values are plotted as dashed lines in Figs. 7(b) and 7(e), where they are seen to give good account for the experiments and simulations.

The predicted values of α and k^* become independent of the size when $R \gg \lambda$, in which case surface tension effects are negligible and $\alpha = \pi - 2\theta_1 = 85^\circ$ and $k^* = 2\pi/\alpha = 4.2$. This suggests that blocky shapes may be the corresponding terminal forms at geophysically relevant scales.

C. Gravitational torque, potential and an evolving stability landscape

The model and simulations provide unique opportunities to investigate the lock-and-flip dynamics as related to the mechanical stability of a buoyant body [37,43]. Specifically, we may assess the rotational stability as dictated by hydrostatic torque and which depends on the shape of the object and its posture while floating at the waterline. The model provides access to such information through a procedure in which we take the computed ice shape at any given time and assess the torque across all possible orientation angles. The assumed posture is defined by the angle $\phi_0(t)$ of an arbitrary reference point that obeys $\dot{\phi}_0 = \omega_{cm}$. For each possible orientation angle ϕ, we solve for the new equilibrium height z_{cm} for which the total force on the body vanishes. Each such posture does not in general satisfy torque balance, and the computed torque $\tau(\phi)$ is purely gravitational since the body is static. The example plot shown in Fig. 10(a) is at a time just after a flip. At this moment, the actual orientation ϕ_0 is near 0 as marked by a filled purple dot, and it has $\tau = 0$ consistent with its equilibrium status. The torque profile $\tau(\phi)$ has negative slope at this point, which implies its stability: small perturbations of either sense are resisted by torques whose signs imply a restoring

effect. The complete profile shows that there are other angles that are similarly stable such as that marked by the solid gray dot, while others are unstable as marked with open symbols.

The stability can be conveniently characterized by defining the potential $\mathcal{U}(\phi) = -\int^{\phi} \tau(\phi')d\phi'$, which in this case represents the gravitational potential energy associated with orientation [59,64–66]. Its minima or valleys signify stable postures, and maxima or peaks indicate unstable postures. The dark curve in Fig. 10(b) corresponds to the same moment shown in Fig. 10(a), and indeed the system sits at a local minimum. The curves above represent earlier moments and those below later moments, and the actual assumed posture is in all cases marked with the filled purple dot. The current posture was therefore arrived at through a transition (arrow) in which a previously stable valley inverted into an unstable peak. Similarly, the current state later gives way to another such transition. These are capsize events, which therefore are triggered by loss of gravitational stability. Their form is consistent with subcritical pitchfork bifurcations that are repeatedly triggered by the changing potential landscape [64].

A more complete picture is provided by the color map of Fig. 10(c) showing the potential across rotation angles and through time. The landscape proves to be corrugated at all moments, and its valleys and peaks change with time. The actual posture assumed is marked by the white dots connected by lines, and the events of Fig. 10(b) are highlighted with the vertical stripe. These data show that the system locks into place at a potential valley (magenta tones) that however gradually increases in energy and eventually becomes a peak (cyan), at which time the system abruptly jumps to a new posture. These jumps are capsize events, and the gradually destabilizing landscape is attributable to the melting-induced changes in shape. Interestingly, not only does the valley of a locked state morph into a peak, but also two adjacent valleys are formed on either side. The system is thus primed to fall into one or the other, corresponding to a flip of either sense and whose characteristic rotation angle is set by the valley locations.

D. Metacenter analysis and shape sensitivity

The rich picture provided by the potential landscape complements the more conventional metacenter analysis of the orientational stability of floating bodies. The metacentric height is defined as $z_{\mathrm{mc}} = z_{\mathrm{cb}} - z_{\mathrm{cm}} + I_{\mathrm{wl}}/A_{\mathrm{b}}$, where $z_{\mathrm{cb}} = \int_{\Omega_{\mathrm{b}}} z\, dA$ is the center of buoyancy computed based on the region Ω_{b} of ice below the waterline, $I_{\mathrm{wl}} = \int_{\mathrm{wl}} (x - x_{\mathrm{cm}})^2\, dx$ is the second moment of area along the waterline, and $A_{\mathrm{b}} = \int_{\Omega_{\mathrm{b}}} 1\, dA$ is the displaced volume of water per unit length [49]. This quantity is related to the linear stability of the equilibrium posture and reflects the growth rate of rotational perturbations. Specifically, positive values imply linear stability while negative values indicate linear instability. Figure 10(d) shows its evolution throughout the simulation. The locked states have $z_{\mathrm{mc}} > 0$ that gradually decreases to zero, at which point a capsize event is triggered and the metacenter quickly jumps to a positive value again. The profile is therefore sawtoothlike. The jumps tend to increase in amplitude over time, suggesting a deepening of the stable valleys that is corroborated by close inspection of the magnitude of $\mathcal{U}(\phi)$ in Fig. 10(c).

Most impressively, the changes in metacentric height are in all cases remarkably small, of order 0.1 to 1 mm as the body progresses through size scales of 1 to 10 cm. This implies that changes in the shape at the 1% level dictate if and when the body loses stability, and these fine-scale effects drive the complex dynamics of capsizing. This extreme sensitivity to shape may explain aspects of the dynamics that vary trial to trial and which differ between experiments and simulations. In particular, the time interval between flips characterized in Fig. 7(f) is seen to vary by an order of magnitude across the experimental runs and simulation. That these variations are attributable to shape sensitivity is further supported by additional simulation runs shown in the Appendix. Small perturbations to the initial shape are shown to lead to wildly different numbers of flips and time intervals. Similarly, changing the curvature smoothing parameter ε in the melting law leads to substantial changes in these factors whereas other aspects such as the flip angle are relatively preserved.

VI. DISCUSSION

Collectively, these findings show how phase change, shape evolution, and gravitational stability are inextricably linked for ice floating and melting in water. The shape-motion dynamics involves a progression to a faceted geometry and lock-and-flip motions that, while complex, can be understood via the presented methods and analysis of the interactive effects. Melting is inherently differential and dominant along the lower, submerged portions, which causes the body to gradually approach gravitational instability, at which point the ice rolls over and again assumes another posture of temporary stability. This recurrent process patterns the body into a quasipolygonal form whose corners are scars of its current and past intersections with the waterline. The system has the mathematical structure of a dynamical system for the rigid-body motions that is driven by and feeds back on the moving-boundary problem for melting. The latter involves gradual reshaping that typically leads to only small changes in posture, and these periods of stasis are punctuated by abrupt and rapid capsize events. Each such event follows the structure of a pitchfork bifurcation, and the system repeatedly returns to these unstable tipping points. These aspects are well summarized by a corrugated potential landscape that is dynamic, with the faceted geometry leading to peaks and valleys that invert due to melting and thereby cause abrupt transitions.

These results provide validation data for future studies aimed at understanding the dynamics of ice at fundamental scales. In particular, application of computational fluid dynamics methods must handle the flow-structure interaction aspect of the rigid-body motion as well as the phase change and hence moving boundary aspect [53,67–70]. The three-phase boundary at the waterline also presents a challenge [71,72], and its effect is shown here to be critical to the shape-flip dynamics. Methods applied to this problem must also account for the impressively wide range of spatiotemporal scales. The featureless initial geometry develops facets and corners that span orders of magnitude in the local shape curvature, and miniscule migrations of the metacentric height at the 1% level drive the complex capsizing dynamics. The ice-water boundary recedes at speeds of order 1 mm/min $=$ 10^{-3} cm/s due to melting and as high as 10 cm/s during rolling.

The systems studied here are too idealized to expect direct relevance to icebergs and natural ice generally. However, the methods are versatile enough to address more naturalistic situations, and our findings suggest interpretations to be tested and further effects to be investigated. The widely varying size scales, water temperatures, and salinity for natural melting are expected to substantially affect the flows, rates, and shape evolution. However, room-temperature fresh water has been viewed as geologically informative even for marine icebergs given knowledge of how environmental conditions map to bulk melting rates [18]. Further, the success of the uniform-rate melt model presented here suggests that the phenomena are dominantly caused by the stark difference in rate for portions above and below the waterline, a feature that can be expected generally. The effect of scale can be accounted for by the size dependence appearing in the melting law proposed here. Whereas capsizing occurs spontaneously via melting in our system, the additional forcings in nature can be expected to prematurely trigger rollover of marginally stable icebergs [36–38]. Regarding stability, it seems a basic lesson learned here is that simplistic shapes (e.g., rectangular cross-section) and the associated approximate stability metrics (critical height-width aspect ratio) may be inadequate as a general rule [16,18,31]. Shape matters, and fine-scale details of the shape can have outsized importance in determining if and when capsize occurs.

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All authors contributed to designing the project, conducting the experiments, analyzing the data, interpreting the results, developing the model, implementing the simulations, and composing the manuscript.

The authors declare no competing interests.

DATA AVAILABILITY

The data that support the findings of this article are openly available [73].

APPENDIX: ADDITIONAL TRIALS AND DATA AVAILABILITY

1. Additional experimental trials

We conducted a total of 10 experimental trials, three of which are presented in Figs. 11(a)–11(c) with before and after shapes shown for every other flip. Comparison among each other and to the data set of Fig. 3 show that the detailed time course of the shape and flip dynamics differ trial to trial while the key aspects and major trends emphasized in our study are well reproduced. We interpret these findings as indicating that the details are sensitive to uncontrolled factors in the experiments such as small differences in the initial shape, while the general outcomes are robust.

2. Additional simulation trials

To further assess the robustness of our findings, we conduct additional simulations for perturbed initial shapes. The computed outcomes for three cases are shown in Figs. 11(d)–11(f), where the perturbations take the form of low-amplitude modulation with shape modes $k_0 = 3$, 5, and 7, respectively. All trials take about the same time to melt down, and all show lock-and-flip dynamics and the emergence of a quasipentagonal form at later times. The number of flips, when they occur, and their distribution over time are less consistent. Among other differences, the $k_0 = 3$ case takes a long time to reach the first flip, and the $k_0 = 5$ case has fewer flips throughout. We interpret these results as indicating that indeed the dynamical details may be sensitive while the general aspects emphasized in our study are more robust. In particular, the quasipentagonal terminal shape emerges from the dynamics in all cases and thus is not a product of the initial condition.

We present in Fig. 12 the results of further simulation runs at different values of the effective diffusivity parameter ε, which dictates the strength of the curvature-smoothing effect in the melt law Eq. (7) [30]. We consider increasing and decreasing the value of this parameter by a factor of 2 relative to the value of $\varepsilon = 5 \times 10^{-4}$ cm^2/s used throughout our work. For all cases, we observe similar lock-and-flip dynamics that yield a quasipentagonal terminal shape. As expected, higher values of ε are associated with smoothing of the corners. These shape details tend to strongly affect the number of flips and the time intervals between flips.

3. Data availability

For each experiment and simulation, we release time-stamped boundary data as well as preprocessed data for the results of this paper. We have made these data publicly available at as a repository on the GitHub site [73]. In the folder `data`, boundary data are contained in subfolders for each of 10 experiments labeled the date `year-month-day`. These folders contain `X.mat` files, where `X` is the time stamp of the frame measured in seconds. Each experiment has an initial $t = 0$ data point, as well as a frame a few seconds before and after each flip. Each `X.mat` file contains `x_true`, `y_true`, `x_interp`, `y_interp`, `cx`, `cy`, and `curvature`, which are rescaled to be in centimeters. Boundary data is aligned horizontally around $x = 0$ and vertically such that the waterline is at $y = 0$. The data arrays are defined as: `x_true` and `y_true` are the unprocessed x and y boundary coordinates of the boundary; `x_interp` and `y_interp` are the x and y coordinates of the Fourier-interpolated and equispaced boundary; `cx` and `cy` are the center of mass coordinates; and `curvature` is the local curvature of the Fourier-interpolated points `x_interp` and `y_interp`, computed as a function of arclength defined counterclockwise.

In the subfolder `experimental-measured-quantities`, we include all experimental preprocessed data included in this paper: area, shape mode, rotation angle, flip times, and flip directions. Data for area and shape mode calculations are taken at $t = 0$ and a few seconds after each flip. For `Area`, `shapeMode`, and `rotationAngle`, the first column is the time stamp in seconds, and

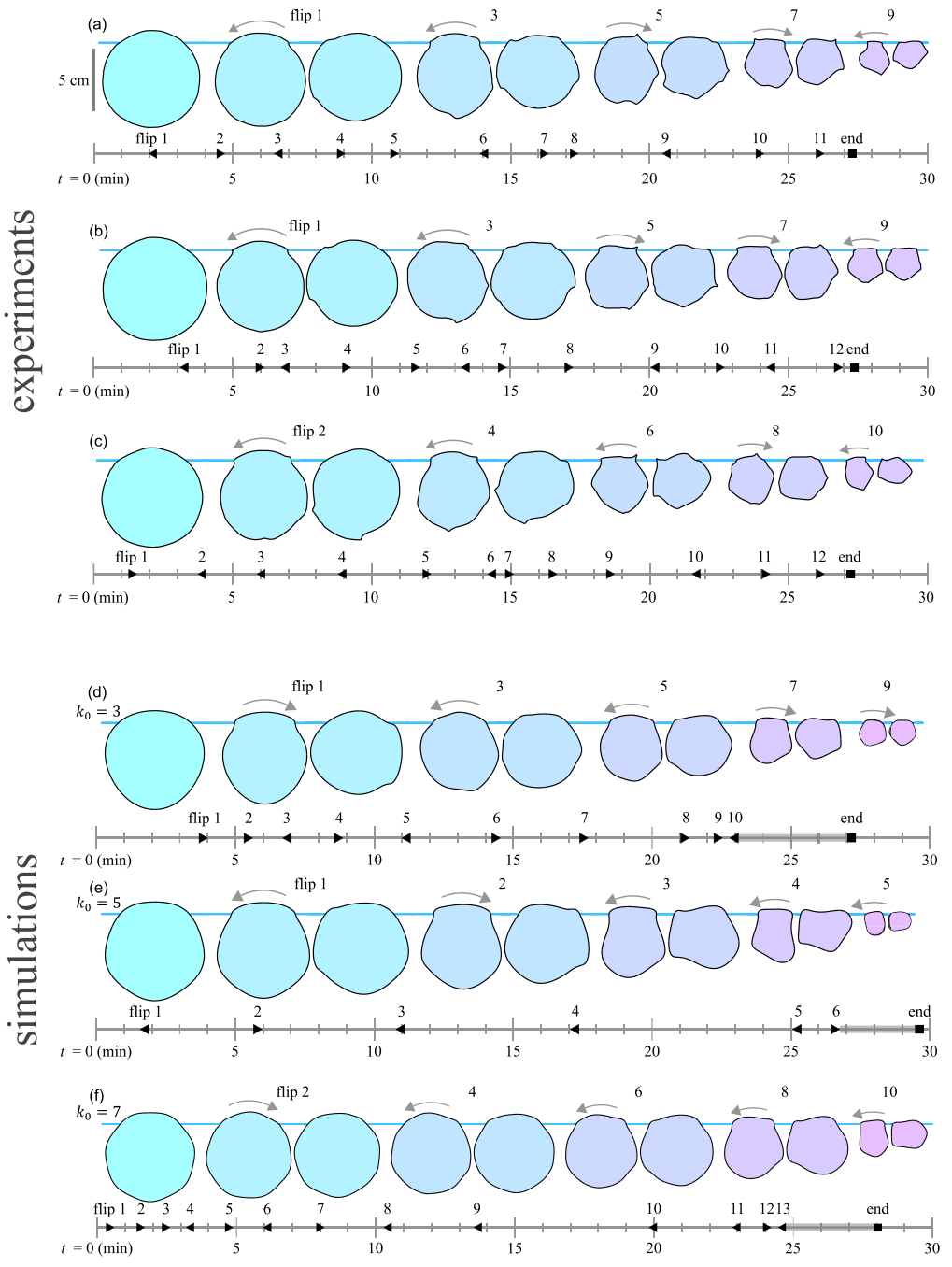


FIG. 11. (a)–(c) Three additional experimental trials. Extracted cross-section shapes at the beginning and before and after selected flips, with flip timelines below. (d)–(f) Simulation results for perturbed initial shapes. Boundary shapes and flip timelines for initial shapes modulated with modes $k_0 = 3$, 5, and 7.

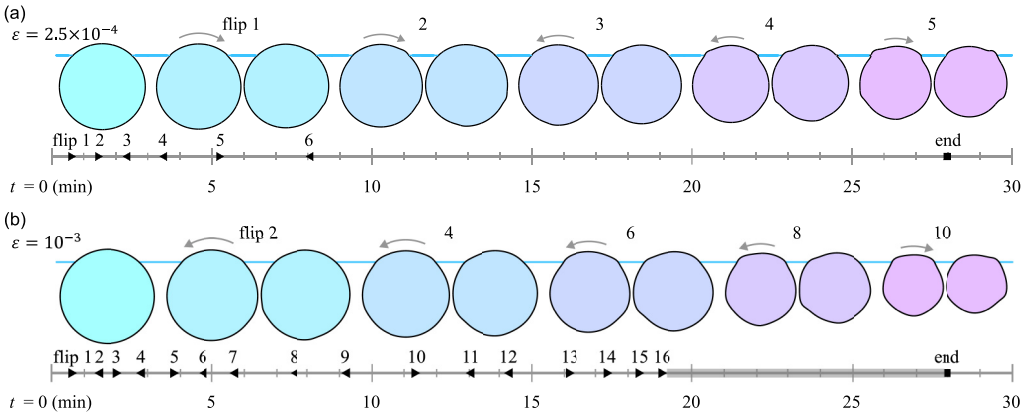


FIG. 12. Shape and capsize dynamics of simulations for which diffusivity parameter is varied relative the value $\varepsilon = 5 \times 10^{-4}$ cm^2/s used throughout this work. (a) Results for ε decreased by a factor of 2 relative to the cases studied in the paper. (b) Results for ε increased by a factor of 2. Higher ε tends to smooth the sharp features, to which the number of flips is sensitive.

the second column is the computed value. `Area` is rescaled to be in centimeters. A positive angle corresponds to a counterclockwise rotation. The array `flipTimes` contains the time in seconds at which all flips occur, and the array `flipDirections` is $+1$ for a counterclockwise flip and -1 for a clockwise flip.

The subfolder `simulationData` contains analogous simulation data, where `simulationBoundaries.mat` is a variable `XX` of boundary data every five seconds. The first column is the grid point, the second the dimension, and the third the time stamp in seconds. The file `simulationData.mat` contains the corresponding extracted quantities taken once per minute and where the area data are normalized by its initial area.

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